Sylvain **Delahaies**

Data Science | Applied Mathematics

In brief

Mathematician, with a strong experience in the field of data assimilation and data analysis, from theoretical aspects to practical implementations, I am immediately available to attend to your needs.

Civil status

Born the 7/10/1979, Age 43

Civil union, 2 children

Contact

sdelahaies@gmail.com

on request

6 impasse du chapial
24570 Condat sur Vézère, France

website: sdelahaies.github.io

Languages

- French - Mother tongue
- English - professional proficiency
- German - Notions

Soft skills

Creativity | Curiosity | Adaptability | Autonomy | Planning | Team working | Problem solving | Communication

Interests

- City council
- Sciences
- Technologies
- Piano, Guitar
- Lutherie
- Computing
- Linux
- craft building
- Hiking

References

- I. Roulstone, Prof., University of Surrey
- D. Lloyd, Prof., University of Surrey
- S. Roudier, Mayor of Condat-sur-Vézère

EDUCATION

2005-2008

PhD — Guildford, UK
University of Surrey
Geophysical Fluid dynamics
Thesis: Contact and Complex structures in Geophysical Fluid Dynamics.

2004-2005

MSc — Guildford, UK
University of Surrey
HydroInformatics

1999-2004

MSc-BSc — Angers, Fr
Université d'Angers
Mathématiques

EXPERIENCE

2020-2022

Visiting research fellow — Guildford,UK
University of Surrey
Department of Mathematics

2009-2020

PostDoc researcher — Guildford,UK
University of Surrey
Department of Mathematics

- Development of bayesian ensemble filters for point processes. Application to crime modeling, neural networks, covid19 outbreak.
- Using insurance claims history and LiDAR elevation data together with machine learning techniques we evaluated insurance risks associated with land subsidence phenomenon. Partnership with the National Physics laboratory (NPL).
- In collaboration with Earthi (UK), preliminary study to evaluate the benefits of using very high resolution satellite imagery together with a terrestrial ecosystem carbon cycle model to monitor and detect forest degradation.
- Analysis of calls for services data from the Providence Police Department to optimize shift scheduling and police patrols dispatch.
- Within the National Centre for Earth Observation (NCEO), development of variational tools to evaluate the information content of different observation fluxes in carbon cycle models.

COMPUTING SKILLS

Languages: Python ●●●, Matlab ●●●, C ●, HTML ●, R ●, bash ●, Fortran ●●, kotlin ●, LaTeX ●●●

Python: munpy ●●●, matlplotlib ●●●, pandas ●, sklearn ●, numba ●, dash ●, pyopencl ●●, tensorflow ●

Softwares: Spyder ●●●, Matlab ●●, Visual code ●, Rstudio ●, Android studio ●, Wordpress ●, Libre/MS Office ●●, Git ●